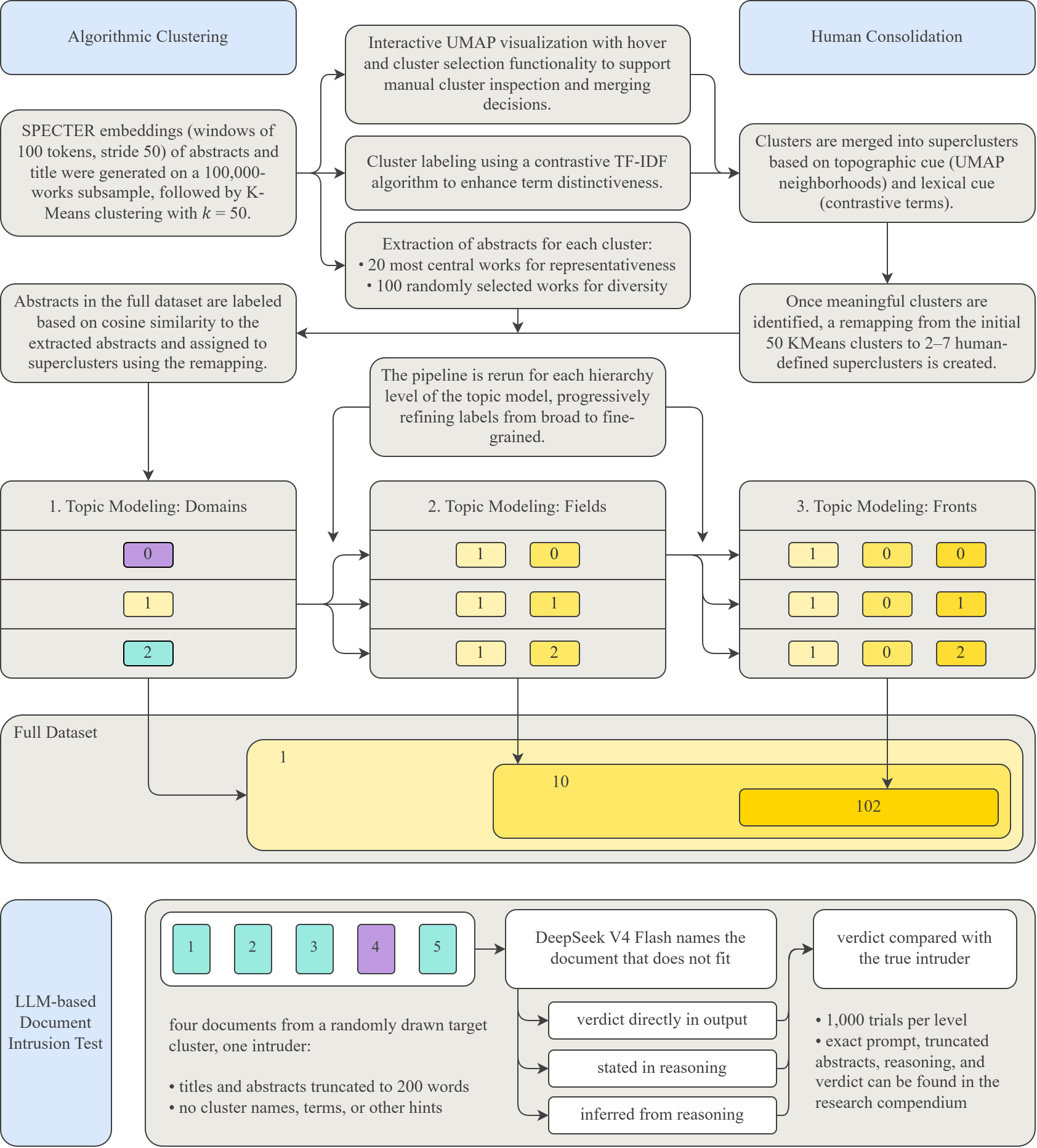

Algorithmic Clustering
Human Consolidation
Interactive UMAP visualization with hover and cluster selection functionality to support manual cluster inspection and merging decisions.
SPECTER embeddings (windows of 100 tokens, stride 50) of abstracts and title were generated on a 100,000-works subsample, followed by K-Means clustering with $k = 50$.
Cluster labeling using a contrastive TF-IDF algorithm to enhance term distinctiveness.
Clusters are merged into superclusters based on topographic cue (UMAP neighborhoods) and lexical cue (contrastive terms).
Extraction of abstracts for each cluster:
• 20 most central works for representativeness
• 100 randomly selected works for diversity
Abstracts in the full dataset are labeled based on cosine similarity to the extracted abstracts and assigned to superclusters using the remapping.
Once meaningful clusters are identified, a remapping from the initial 50 KMeans clusters to 2–7 human-defined superclusters is created.
The pipeline is rerun for each hierarchy level of the topic model, progressively refining labels from broad to fine-grained.
1. Topic Modeling: Domains
0
1
2
2. Topic Modeling: Fields
1 0
1 1
1 2
3. Topic Modeling: Fronts
1 0 0
1 0 1
1 0 2
Full Dataset
1
10
102
LLM-based Document Intrusion Test
1 2 3 4 5
four documents from a randomly drawn target cluster, one intruder:
• titles and abstracts truncated to 200 words
• no cluster names, terms, or other hints
DeepSeek V4 Flash names the document that does not fit
verdict directly in output
stated in reasoning
inferred from reasoning
verdict compared with the true intruder
• 1,000 trials per level
• exact prompt, truncated abstracts, reasoning, and verdict can be found in the research compendium